RU Financial Services

Vision

Version 1.5

© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

Revision History

Date	Version	Description	Author
19th January, 2000	1.0	First draft	J. Carlos Goti
27th January, 2000	1.0	First complete draft	J. Carlos Goti
17th November, 2000	1.2	Revision for clarification	J. Bell
16th October, 2001	1.3	Updated for new release	B. Baker
30th October 2001	1.4	Editorial changes	B. Baker
12th February 2003	1.5	McKinley updates	B. Baker

© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

Table of Contents

1. Introduction 5
   - 1.1 Purpose 5
   - 1.2 Scope 5
   - 1.3 Definitions, Acronyms, and Abbreviations 5
   - 1.4 References 5
   - 1.5 Overview 5

2. Positioning 5
   - 2.1 Business Opportunity 5
   - 2.2 Problem Statement 6
   - 2.3 Product Position Statement 7

3. Stakeholder and User Descriptions 8
   - 3.1 Stakeholder Summary 8
   - 3.2 User Summary 8
   - 3.3 User environment 9
   - 3.4 Stakeholder Profiles 9
     - **3.4.1 RU e-st** Product Manager 9
     - **3.4.2 RU e-st** Business Manager 10
     - **3.4.3 RU e-st** Development Manager 10
     - **3.4.4 RU e-st** Partner Executive 10
   - 3.5 User Profiles 11
     - **3.5.1 RU e-st** End User 11
     - **3.5.2 RU e-st** System Administrator 11
   - 3.6 Key Stakeholder/User Needs 12
   - 3.7 Alternatives and Competition 12
     - 3.7.1 Stay with the current system. 12
     - 3.7.2 Buy an existing online securities trading service company. 12
     - 3.7.3 Buy an off-the-shelf securities trading system and install it. 13

4. Product Overview 13
   - 4.1 Product Perspective 13
   - 4.2 Summary of Capabilities 14
   - 4.3 Assumptions and Dependencies 14
   - 4.4 Cost and Pricing 14
   - 4.5 Licensing and Installation 14

5. Product Features 15

6. Constraints 15

7. Quality Ranges 15

8. Precedence and Priority 15

9. Other Product Requirements 16
   - 9.1 Applicable Standards 16

© Copyright IBM Corp. 2003, 2010

Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

9.2 System Requirements 16
9.3 Performance Requirements 16
9.4 Environmental Requirements 16

10. Documentation Requirements 16
10.1 User Manual 16
10.2 Online Help 16
10.3 Installation Guides, Configuration, Read Me File 16
10.4 Labeling and Packaging 16

11. Appendix 1 - Feature Attributes 16

© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

RU e-st Project Vision

1. Introduction

The purpose of this document is to collect, analyze, and define high-level needs and features of the RU e-st system. It focuses on the capabilities needed by the stakeholders, the target users, and why these needs exist. The details of how RU e-st fulfills these needs are shown in the use case and supplementary specifications.

This document describes the vision for the proposed RU e-st system. The RU e-st system allows its users to trade securities online, using a Web interface (an existing browser). It enables its users to perform the basic operations: open accounts, trade, and obtain information about their accounts and about what is happening in the securities markets.

1.1 Purpose

The objective of writing a Vision document for the RU e-st system is to enable agreement among stakeholders, developers, and business representatives. Another purpose of writing a Vision document is to provide a common platform for agreement between the developers themselves.

1.2 Scope

This document records the vision for the RU e-st system. It is associated with a project to provide Web-based online trading facilities through financial institutions.

1.3 Definitions, Acronyms, and Abbreviations

Definitions of terms and allowed operations are to match the standard ones in use in the field of securities trading. See the Glossary document for details.

1.4 References

[1] RU Financial Services RU e-st Requirements Management Plan
[2] RU e-st Project Glossary

1.5 Overview

This document contains product positioning statements, an analysis of the system's stakeholders, an analysis of the expected system users, and a list of features that the system should deliver. These features are derived from the input obtained from stakeholders.

2. Positioning

2.1 Business Opportunity

Currently, RU Financial Services has a number of critical business objectives, which, if not met, threaten the long-term viability of the organization. The success of this project will enable RU Financial Services to continue to grow and move into the 21st century from a position of strength.

© Copyright IBM Corp. 2003, 2010

Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

2.2 Problem Statement

The problem of	a stock trading system that: • Lacks features required by our customers. • Has intermittent errors in trades. • Is expensive to operate and maintain due to labor and infrastructure costs. • Is difficult to maintain due to: ○ Client/server architecture. ○ Legacy code that lacks documentation, structure and architecture. ○ Lack of developer knowledge of current system.
affects	RU Financial Services.
The impact of which is	We need to replace the current online stock-trading system.
A successful solution would	• Provide the feature set demanded by our customers. • Provide a reliable service that has no errors in trades. • Reduce the operation and maintenance costs to no more than 10% of the gross contribution margin of the online trading business. • Use a corporate standard architecture and implementation language.

© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

2.3 Product Position Statement

For	RU Financial Services
who	needs to replace the current stock trading system because it is losing profitability, customers, and market share in the e-stock trading market.
The RU e-st	is a Web-based securities trading system
that	has the full stock trading functionality demanded by our customers.
Unlike	continuing with the current client-server-based system that is: • Running on unsupported hardware. • Is difficult and expensive to maintain. • Lacks some of the features required by our customers. • Is prone to error. • Requires our customers to install client software on their machines.
Our product	will make the business profitable again by: • Providing error free trading that provides all of the most popular features requested by our customers. • Allowing bug fixes and upgrades to be applied quickly and cheaply due to its Web-based architecture. • Reducing operating costs by using supported hardware and using standard maintenance contracts. • Reducing maintenance costs by using corporate standard programming languages and architecture.

© Copyright IBM Corp. 2003, 2010

Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

3. Stakeholder and User Descriptions

3.1 Stakeholder Summary

There are typically a large number of stakeholders to a project. Here we list the ones that have primary responsibilities and have the largest influence on the success of the project.

Name	Represents	Role
RU e-st Product Manager	Represents the executive management in the company.	Define a product that satisfies the business requirements. Evaluate options and scope the different versions of the system.
RU e-st Business Manager	The business entity that makes the funding decisions for development.	Assure that the company has the plans and resources to deliver the system and that it is a good business decision for the company to spend the funds.
RU e-st Development Manager	The team that develops and releases the system.	Assure that the development team delivers a system that complies with the specifications supported by the RU e-st Product Manager.
RU e-st End User	People who trade securities using the system.	Validate that the features offered by the system are adequate and responsive to the needs of traders.
RU e-st Partner Executive	Companies whose systems support the operation of the system. They are: Market Trading, News System, and Quote System.	Approve business agreements for specialized functions needed to support online securities trading.

3.2 User Summary

Name	Description	Stakeholder
End User-Day Trader	A very knowledgeable end user, who is on the system very frequently and is aware of and makes use of advanced features.	RU e-st End User
End User-Naïve	First-time or infrequent end user of the system. This person needs guidance and might need to refer to online help for definitions and business consequences of trading actions.	RU e-st End User

© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

RU e-st Administrator	Does customization and setup activities on the system. Maintains up-to-date information and produces and analyzes operational reports of system performance.	RU e-st Development Manager
RU e-st Support	Helps calling customers use the system correctly. Reports errors and develops operational fixes.	RU e-st Development Manager
RU e-st Developer	Part of the team that develops the system (software developer, tester, etc.)	RU e-st Development Manager

3.3 User environment

End users typically work alone. They have logged on to the Internet (hotel room, airport, and so on) and want to check the status of their investment or perform one or more trades.

Except for the people who share an account, there is no interplay between the different end users. They are independent of each other.

It is expected that many thousands of end users will be trading at the same time.

End user accesses the system through a standard Internet browser. They do not need to learn new user interface (UI) techniques of any type. They access the system from many different client platforms (Windows, McIntosh, Linux, OS/2).

3.4 Stakeholder Profiles

3.4.1 *RU e-st Product Manager*

Representative	The online trading business
Description	Defines a product that satisfies the business requirements. Evaluates options and scope the different versions of the system.
Type	Is an expert-level user of the system.
Responsibilities	Does the business analysis of the system for justification to the RU e-st Business Manager. Achieves a definition of the system that can be built and delivered on time and within cost. Commits the marketing and sales resources to make money with the product.
Success Criteria	The product (system) is accepted and makes money, as evidenced by a continued trend of reductions in monthly loss of customers with a corresponding increasing trend in monthly profits.
Involvement	Constantly reviews plans and development progress. Assesses how features are being supported. Validates expenses by system feature.
Deliverables	Participates during iteration assessment sessions and when project decisions are made.
Comments/Issues	Needs to keep an eye on the competition to ensure success in the marketplace.

© Copyright IBM Corp. 2003, 2010

Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

3.4.2 *RU e-st Business Manager*

Representative	The business entity that makes the funding decisions for development.
Description	Ensures that the company has the plans and resources to deliver the system and that it is a good business decision for the company to spend the funds.
Type	Probably not a user of the system.
Responsibilities	Assures the funds are committed to the project and adjustments are made with sound business sense. Influences marketing and sales to prioritize the system according to the company's vision. Promotes the system with potential customers.
Success Criteria	The system makes money for the company and is aligned with the overall strategy. The company gets good press from the system, and the stock goes up.
Involvement	Participates in the system's phase assessments.
Deliverables	Provides the project team with resources and promote the system vision with good press inside and outside the company.
Comments/Issues	The RU e-st Business Manager should be supported with solid marketing and positioning information about the system.

3.4.3 *RU e-st Development Manager*

Representative	Of the team that develops and releases the system.
Description	Ensures that the development team delivers a system that complies with the specifications supported by the RU e-st Product Manager.
Type	Very proficient in the use of the system.
Responsibilities	Allocates the resources and provides the technical and business controls to assure the success of the project.
Success Criteria	The project completes a sound delivery that can be sold, as specified by the RU e-st Product Manager.
Involvement	The Development Manager works closely with the Project Manager, the Architect, and the Analyst to ensure that all is on the right track.
Deliverables	Supports the project with controls, imaginative resolution of problems, allocation of resources, and teams building activities.
Comments/Issues	The Development Manager can act as a technical leader in some areas of expertise.

3.4.4 *RU e-st Partner Executive*

Representative	Companies whose systems support the operation of RU e-st. They are: Market Trading, News System, Quotes System, and the Financial Network System.
Description	Approve business agreements for specialized functions needed to support online securities trading.
Type	Might not use the system at all.
Responsibilities	Achieves business agreements for cooperation between systems. Agree on business synergism and mutual charging schemes.
Success Criteria	Smoothes business operation (low bureaucracy) during the system's joint operations.
Involvement	Early agreements during the inception phase of the system. The business arrangements should be completed early in the development process.
Deliverables	Mutual service support during the operation of the systems. Clear arrangements for

© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

	production problem resolution.
Comments / Issues	Business arrangements are limited in time. Backup arrangements are always good to have. In some cases, we might want to have two or more partners providing the same services to minimize full dependencies.

3.5 User Profiles

Users are listed in two categories: end users and RU e-st system administrators. The breakdown shown above is too detailed for this section.

3.5.1 *RU e-st End User*

Representative	The RU e-st Product Manager is the representative of End Users.
Description	Uses the RU e-st system to perform securities trades.
Type	The Day Trader is very knowledgeable about the system's operations and features because of constant use. The naïve user is either a first-time user or an infrequent user.
Responsibilities	End users give the development team feedback about how the system supports their trading needs. They should point out lacking features and awkward operational characteristics of the system.
Success Criteria	Users benefit from an improved system access to their needs.
Involvement	Users should participate in the description of use cases, the evaluation of partial implementations, and the assessment of the help features of the system.
Deliverables	Users deliver feedback to the team's analysts and architect either verbally or in writing.
Comments / Issues	It is very important to the project to have good user representatives. They can improve the system by sharing their domain expertise and operational needs.

3.5.2 *RU e-st System Administrator*

Representative	System administrators are represented by the RU e-st Development Manager.
Description	System administrators do set up, customization, support and development of the system.
Type	All system administrators are users at the expert level.
Responsibilities	Prepares and executes installation and customization operations. Tends to production problems (errors, performance, work-around solutions, etc.) Supports end users who call for help or to report errors. Develops and test parts of the system.
Success Criteria	Customer satisfaction measures.
Involvement	Should participate during development in the same way that end users do. They should concentrate on the support and maintenance use cases.
Deliverables	Comments to improve the support and maintenance aspects of the system.
Comments / Issues	Often, the system administrator users are not brought on-board in time to improve the system. A special effort should be made to get their feedback as early as possible.

© Copyright IBM Corp. 2003, 2010

Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

3.6 Key Stakeholder/User Needs

Requirements	Origin
STRQ1: Want to be able to transfer funds from other accounts (not necessarily held with this firm) to a trading account.	Trading Customer
STRQ2: State and federal regulations require monthly reports of account activity. Refer to specification RUFS-1234 for details of the information required.	Regulatory
STRQ3: The system should allow the use of any browser.	Trading Customer
STRQ4: Customers want to manage their retirement funds.	Trading Customer
STRQ5: Must be able to upgrade the system without taking it offline.	On-line Trading Business
STRQ6: The system should allow traders to trade in multiple markets across the world.	Trading Customer
STRQ7: Must be able to provide convenient answers to customer's most common questions.	Customer Support
STRQ8: The system must provide a secure environment that prohibits fraudulent access.	Trading Customer
STRQ9: Need a way to train customers in the use of the system quickly and conveniently.	Customer Support
STRQ10: The system must operate on hardware that falls under the company's current maintenance contracts.	Finance Department
STRQ11: Need to be able to maintain the system with our current IT hardware and skills. Refer to enterprise architecture document EA-1234 for details.	System Administration
STRQ12: Need account activity statements for tax reporting.	Trading Customer
STRQ13: The system must provide all the basic capabilities of a normal stock broking firm.	Trading Customer
STRQ14: Need to be able to perform research on any given stock.	Trading Customer
STRQ15: The system must allow traders to obtain up-to-date news and alerts on nominated stock.	Trading Customer
STRQ16: The system must provide current and historical information on Trading Acounts. Such as number of shares held, current price, total Trading Account value	Trading Customer
STRQ17: The system shall provide the following types of trades: Market Trades (buy and sell), Limit Trades (buy and sell), and transfers between mutual funds.	Trading Customer
STRQ18: The system shall use the existing Market Trading System to perform securities trades.	Trading Customer
STRQ19: The system must report yearly tax information to the IRS and state tax boards.	Regulatory
STRQ20: Tax forms will be sent electronically to the IRS and state tax authorities.	Regulatory
STRQ21: Tax information can be viewed on-line by the Trading customer and printed if requested.	Regulatory
STRQ22: The system performance must be aceptable to customers that do not have high-speed data access for their computers.	Trading Customer

3.7 Alternatives and Competition

3.7.1 Stay with the current system.

This is not a viable alternative due to the current operating costs and downward trend in customer base.

3.7.2 Buy an existing online securities trading service company.

RU Financial Services could buy an existing online trading company and merge with it. This is surely a possibility, but there are more institutions wanting to buy than the number of companies in a position to be bought. Most companies attempting to make this move need a different alternative.

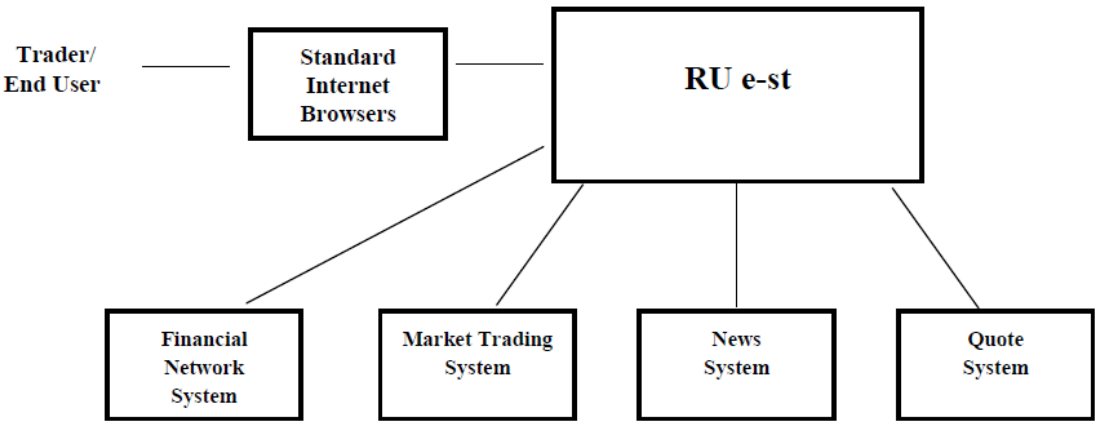
© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

3.7.3 Buy an off-the-shelf securities trading system and install it.

This option is attractive because it eliminates development risks and produces results very quickly. Sourcing an adequate solution is difficult as there are not many commercially available stock-trading products available at this stage.

4. Product Overview

RU e-st is planned to interface with other systems in the following fashion:



All end users access the system's features through standard Internet browsers. The system provides facilities for traders to set up accounts, get securities quotes, obtain news headlines about specific securities, and perform trades. RU e-st keeps information about user's accounts and the assets in them.

4.1 Product Perspective

RU e-st interacts with other systems for some specific tasks. They are:

- **Market-Trading System**: to complete trade orders as specified by the trader.
- **News System**: to retrieve headline news and the corresponding links to full articles for a given security.
- **Quote System**: to retrieve current and historical quote information for a given security.
- **Financial Network System**: to perform money transfers to and from the customers' financial institution accounts.

© Copyright IBM Corp. 2003, 2010

Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

4.2 Summary of Capabilities

RU e-st System	
Trader Benefit	**Supporting Features**
Trading online	• Online trading • Online account setup • Online quotes • Online news
Securities research online	• Online quotes • Online news • Online account reports
Portfolio management online	• Online account setup • Online accounts reports • Online transfers between accounts • Online transfers with banks
Portfolio record keeping	• Historical account reports • Income tax account reports • IRS reporting forms

4.3 Assumptions and Dependencies

The basic features of the system are attainable with today's technologies. No high-risk dependencies are envisioned.

4.4 Cost and Pricing

Marketing is working on the pricing points.
Development is working on the cost estimates.

4.5 Licensing and Installation

Not applicable.

© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

5. Product Features

Features
FEAT1: The system uses the Financial Services Network to enable transfer of funds between other financial institutions and the customers Trading Account.
FEAT2: The system provides automatic reporting of tax-related information to the IRS and state tax boards.
FEAT3: The system supports the most common browsers: Microsoft Internet Explorer, (version 4 and higher) and Netscape Navigator, (versions 4.72 through 4.75 and 6.2).
FEAT4: The system allows funds to be transferred to and from Mutual Fund accounts.
FEAT5: The system allows a customer to operate multiple Trading Accounts with a single login.
FEAT6: The system can be upgraded without taking services off-line.
FEAT7: During upgrades the system will preserve all "in-progress" transactions to ensure an error free experience for the customer.
FEAT8: The system uses the Market Trading System to allow trading in any market worldwide.
FEAT9: The system provides an FAQ page to answer customer's most frequent questions quickly and simply.
FEAT10: The system provides an "I need help now" capability that will open a "chat window" with the next available customer support representative. If there is no support personal available the system will inform the customer where they are in the queue for help.
FEAT11: All Web pages shall be encrypted using 128 Bit SSL Security.
FEAT12: All personal and financial information shall be stored on a separate system with a secure network connection between the systems.
FEAT13: The system provides comprehensive "how-to" documentation that is structured like a story of using the system for a particular purpose. For example, "How to perform a Limit Buy."
FEAT14: The system runs on a corporate standard platform.
FEAT15: The system provides electronic and printed statements of account activity, including profit and loss information, for yearly income tax reporting.
FEAT16: The system provides statements of Trading Account activity including transfers between Trading Accounts, Trade summaries including: ticket code, number of shares traded and trade price.
FEAT17: Transfer cash from one customer trading account to another.
FEAT18: The system executes transactions in an Internet comfortable speed (less than 3 seconds, not counting transmission times on a 56K modem).
FEAT19: Charge to a credit card and place funds in a customer's trading account.
FEAT20: Rollover from a retirement account in another institution in to a retirement account managed by the system.
FEAT21: The system allows Market Buy and Sell transaction types.
FEAT22: The system allows Limit Buy and Sell transaction types.
FEAT23: The system allows mutual fund management.
FEAT24: The system allows stock research and alerts - online quotes, news flashes, and so on.

6. Constraints

All the online features must run on the selected Internet browsers.

Need to perform an additional assessment by the end of the elaboration phase.

7. Quality Ranges

Refer to expected industry standards in this area. Further research is needed.

8. Precedence and Priority

This information is included in the RequisitePro database. Appropriate reports are designed to convey this information.

© Copyright IBM Corp. 2003, 2010

Course materials may not be reproduced in whole or in part without the prior written permission of IBM.

9. Other Product Requirements

9.1 Applicable Standards

Refer to standards related to the versions of HTML and Java by the supported Internet browser.
Refer to standards for financial transactions between institutions in the Financial Network.
The application must conform to all the rules of the NASD, SIPC, and other trading institutions involved in transactions.

9.2 System Requirements

The customer needs a computer (client) with capabilities to access the Internet. Access speed and local hard drive use depends on the customer's configuration and practices of use.

9.3 Performance Requirements

Transactions (except for trades, which require a sale or buy operation) must execute in an Internet-comfortable speed (less than 3 seconds, not counting transmission times).

9.4 Environmental Requirements

No additional requirements other than the ones in common practice for a computer with Internet connection capability.

10. Documentation Requirements

Basically, all documentation for the RU e-st system is available online and not in print form.

We might need some printed information at the brochure level for mailings and "getting started" help. This information should describe trading terms and conditions and a basic description of capabilities.

10.1 User Manual

User manuals shall be available in electronic format and can either be downloaded in PDF form or as a series of HTML pages that can be browsed.

10.2 Online Help

The system offers extensive online help.

The help has two basic objectives:
Assistance in the operation of the system
Assistance with the domain (definitions, descriptions of accounts, transactions, legal issues, etc.)

10.3 Installation Guides, Configuration, Read Me File

No installations are required. No need for this documentation.

10.4 Labeling and Packaging

There is no applicable labeling and packaging activity.

11. Appendix 1 - Feature Attributes

Refer to [1].

© Copyright IBM Corp. 2003, 2010
Course materials may not be reproduced in whole or in part without the prior written permission of IBM.